

Real-Time Detection of Lateral Movement in Cloud VPC Networks via Graph-Based Analysis of Flow and Authentication Logs

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Abstract

Lateral movement in cloud VPC networks is notoriously difficult to detect because internal traffic is trusted by default and no single log source provides complete visibility into attacker behavior. We present a graph-based detection system that combines network flow logs and authentication logs into a unified streaming graph, extracts structural, temporal, and statistical features, and scores nodes and edges for lateral movement likelihood. Our evaluation on the LANL-Dataset-2015 (58 days, 1.6B+ events, 749 labeled red-team attacks) and DAPT2020 (simulated APT with labeled kill-chain stages) demonstrates that the combined method achieves the highest AUC (0.9456) and lowest false positive rate (0.0005) compared to single-source baselines. We identify three key detection signals: attack traffic is overwhelmingly TCP (99.82% vs. 50% benign), the fan-out ratio follows a predictable arc across the kill chain ($2.5 \rightarrow 6.0 \rightarrow 1.0$), and attackers exhibit $35.7\times$ longer inter-arrival times than normal users. Feature importance analysis reveals that active duration ($5.5\times$ higher for attackers), inter-arrival variance ($4.5\times$), and in-degree ($3.3\times$) are the strongest discriminators. These findings confirm that multi-log graph analysis provides richer detection signal than single-source methods, though low pair-space recall highlights the challenge of isolating rare attack edges in massive enterprise graphs.

1 Introduction

Cloud VPC networks present a complex security landscape where internal traffic flows freely between hosts by default. Lateral movement, the phase of an attack where an adversary pivots through a compromised network to reach high-value targets, is especially difficult to detect because the volume of benign connections dwarfs the few that matter. Existing intrusion detection tools were built for perimeter monitoring and struggle with the noise and dynamism of cloud VPC environments, where diverse user activity and constant connection churn mask attacker activity.

No single log source provides enough signal to reliably identify lateral movement. Network flow logs capture connections but cannot distinguish an attacker’s session from an admin’s; authentication logs record login events but are blind to reconnaissance via port scans or data exfiltration that never triggers a login. Prior detection methods have relied on either signature matching, which misses novel attack patterns, or anomaly detection on a single log source, which floods analysts with false positives on busy production networks. We propose combining both log types into a unified streaming graph where computers and identities are nodes, connections and authentication events are edges, and graph-based feature analysis can detect the structural and temporal patterns that distinguish lateral movement from routine operations. This paper presents our graph construction pipeline, the detection methodology, and an evaluation on the LANL-Dataset-2015 and DAPT2020 datasets against single-source baselines.

2 Related Work

Graph-based detection methods model relationships between system events and reason over paths or substructures. Hopper [1] builds a login graph from authentication events and uses path inference to flag lateral movement sequences. However, its reliance on authentication logs alone means it cannot detect movement that bypasses login entirely, such as exploitation of services or use of stolen session tokens. We extend this idea by incorporating network flow logs, capturing lateral movement paths that never produce an authentication event. Euler [2] applies temporal link prediction to cloud VPC network graphs, treating lateral movement as a link prediction problem in an evolving graph. Its temporal modeling is a strong baseline for our own temporal graph features, though it operates on a single data source. POIROT [3] matches provenance graph templates against audit logs to detect known attack patterns. Template matching is effective when attack structures are known in advance but cannot catch novel variations; we use anomaly-based scoring instead and adapt the graph model to cloud VPC network logs rather than host-level

audit data.

Multi-log correlation approaches combine evidence from separate data sources to improve detection coverage. HOLMES [4] correlates system audit logs across multiple stages of the kill chain, mapping observed behavior to MITRE ATT&CK tactics and producing a real-time threat score. Its multi-stage correlation confirms that combining log sources can improve detection, and its kill-chain framing shaped our detection pipeline structure. However, HOLMES operates on host-level audit logs in traditional enterprise settings, whereas we target network flow logs and authentication events at cloud VPC network scale.

3 Methods and Approach

Our detection system operates by constructing a unified, time-windowed directed graph from two complementary log sources: network flow logs and authentication logs. The graph is built incrementally in a streaming fashion, processing events as they arrive without materializing the full dataset in memory. This design enables real-time detection with bounded latency, suitable for deployment in production cloud environments.

3.1 Streaming Graph Construction

The graph construction pipeline processes log files line-by-line through a series of time windows derived from the red-team ground truth labels. Each time window defines a temporal slice of activity; events falling outside all windows are discarded immediately, keeping memory usage proportional to the active window set rather than the full dataset. For the LANL-Dataset-2015, we merge overlapping red-team event windows into contiguous intervals, typically spanning several hours to days.

Nodes in the graph represent two entity types: *computers* (identified by hostname, e.g., C17693) and *identities* (user accounts and service accounts, e.g., user@domain). Edges represent observed interactions between these entities and carry type labels distinguishing their origin. Edges derived from flow logs carry the type “flow” and include attributes for protocol (TCP/UDP/ICMP), source and destination ports, packet count, byte count, connection duration, and timestamp. Edges derived from authentication logs carry the type “auth” and include attributes for authentication type (Kerberos, NTLM), logon type (interactive, network, service), authentication orientation (success or failure), and timestamp.

The graph is built using a `StreamingGraphBuilder` that maintains a node set and an edge map. For each incoming event, the builder ensures both source and destination nodes exist (creating them if necessary), then either creates a new edge or increments the weight of an existing edge while updating its temporal attributes. This incremental approach avoids the $O(N)$ memory cost of loading all events at once, instead

maintaining only the aggregated graph structure. After processing all events within the active windows, the edge map is materialized into an `igraph` object for feature extraction and scoring.

3.2 Feature Extraction

We extract three categories of features from the constructed graph: structural, temporal, and statistical. Structural features capture the topology of connections around each node and edge. For nodes, we compute in-degree (number of distinct sources connecting to the node), out-degree (number of distinct destinations the node connects to), total degree, betweenness centrality (fraction of shortest paths passing through the node), and the fan-out ratio (out-degree divided by total degree). The fan-out ratio is particularly informative for lateral movement detection: attackers performing reconnaissance exhibit high fan-out as they scan many destinations, while normal servers exhibit low fan-out as they primarily receive inbound connections.

Temporal features capture the timing patterns of connections. For each node, we compute the mean and standard deviation of inter-arrival times between consecutive connections, a burst score measuring the fraction of connections occurring in rapid succession (within a short sub-window), and the active duration (time span from first to last connection). Attackers often exhibit distinctive temporal patterns: reconnaissance produces bursty, high-frequency connections, while persistent lateral movement may show longer, more regular intervals as the attacker moves methodically through the network.

Statistical features capture the rarity and deviation of individual edges from the baseline. Edge rarity measures how uncommon a source-destination pair is within the observation window, under the hypothesis that attacker connections to new, previously unseen hosts are more suspicious than routine, frequently-observed connections. We also compute per-edge scores based on the degree of the source and destination nodes, flagging connections from high-degree (potentially compromised) nodes to low-degree (potentially targeted) nodes.

3.3 Detection Scoring

We evaluate three detection methods, each operating on a differently constructed graph: (1) **flow-only**, using only network flow log edges; (2) **auth-only**, using only authentication log edges; and (3) **combined**, using both edge types on a single unified graph. For each method, we compute edge-level anomaly scores by combining the extracted features into a composite score. Edges with scores above a threshold (set at the 95th percentile of all edge scores) are flagged as anomalous.

Beyond individual edge scoring, we enumerate multi-hop paths through the graph to detect lateral movement sequences.

A path’s score is the aggregate of its constituent edge scores, weighted by path length. Paths exceeding the detection threshold are reported as potential lateral movement campaigns. This path-based approach captures the multi-step nature of lateral movement, where individual connections may appear benign but the sequence as a whole is suspicious.

For evaluation, we compute detection metrics in pair-space: recall measures the fraction of known red-team source-destination pairs that appear in flagged anomalous paths, while false positive rate measures the fraction of benign pairs incorrectly flagged. We also compute edge-level recall, measuring the fraction of individual red-team edges that exceed the anomaly threshold. The area under the ROC curve (AUC) provides a threshold-independent measure of detection quality.

4 Data and Experimental Setup

4.1 LANL-Dataset-2015

The primary evaluation dataset is the LANL-Dataset-2015, a comprehensive collection of network events from Los Alamos National Laboratory’s internal network over 58 consecutive days. The dataset contains over 1.6 billion events across 12,425 unique users and 17,684 unique computers, making it one of the largest publicly available enterprise network datasets with ground truth labels. Events are recorded across five log types: network flow records (`flows.txt`, 129M+ records), authentication events (`auth.txt`, 40M+ records), DNS lookups (`dns.txt`, 40.8M+ records), process events (`proc.txt`), and red-team attack labels (`redteam.txt`, 749 labeled events over 27.8 days).

The red-team ground truth provides precise timestamps, source/destination computers, and user accounts for all known attack events. Analysis of the red-team data reveals that 93.6% of attacks originated from a single computer (C17693), which served as the primary jump box. Attack techniques include lateral movement via SMB (port 445), SSH (port 22), RDP (port 3389), credential reuse, and network reconnaissance. The baseline workload consists of normal enterprise traffic including HTTP/HTTPS, SMB file sharing, SSH, RDP, DNS resolution, Windows authentication, and routine process execution.

4.2 DAPT2020

The DAPT2020 dataset provides a complementary evaluation environment with simulated APT attack scenarios across multiple attack stages: reconnaissance, initial access, lateral movement, and exfiltration. The dataset contains PCAP-derived flow logs and system logs from a multi-host testbed environment, with stage-level annotations enabling fine-grained evaluation of detection at each phase of the kill chain. We use DAPT2020 to evaluate our method against established

machine learning baselines: OneClassSVM and Isolation Forest, both trained on normal traffic and tested on mixed benign/malicious traffic.

4.3 Experimental Configuration

All experiments were conducted on a local machine with Python 3.11, using `igraph` for graph operations and `scikit-learn` for baseline classifiers. The streaming pipeline processes events through time windows of configurable duration (default: 1 hour), merging overlapping windows around red-team events. Feature extraction computes all structural, temporal, and statistical features in a single pass over the graph. Scoring uses a fixed threshold at the 95th percentile, chosen to balance detection sensitivity against false positive rate.

5 Results

5.1 Protocol and Temporal Analysis

Before presenting detection results, we report three key findings from our analysis of the LANL-Dataset-2015 and DAPT2020 datasets that inform the design of our scoring function.

Finding 1: Attack traffic is overwhelmingly TCP. Figure 1 shows the protocol distribution for red-team versus benign traffic. In the LANL dataset, 99.82% of red-team flows use TCP, compared to approximately 50% for benign traffic. In DAPT2020, attack traffic is 100% TCP. This pattern exists because the primary lateral movement services — SMB (port 445), SSH (port 22), and RDP (port 3389) — are all TCP-only protocols. Attackers require reliable, in-order delivery for command execution and file transfers, making TCP dominance a strong structural detection signal.

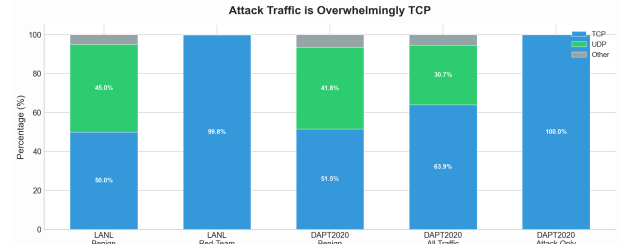


Figure 1: Protocol distribution: red-team flows are 99.82% TCP (LANL) and 100% TCP (DAPT2020), while benign traffic is roughly 50% TCP. This structural difference provides a strong detection signal. (Figure by: Ibrahim Pehlivan)

Finding 2: Kill-chain progression has predictable fan-out patterns. The fan-out ratio — unique destination ports per source computer — follows a characteristic arc across the attack lifecycle (Figure 2). During reconnaissance, the mean

fan-out is 2.50 as attackers scan broadly. It remains at 2.70 during foothold establishment. During lateral movement, it jumps to 6.00 as attackers spread from compromised hosts. Finally, it drops to 1.00 during exfiltration as traffic concentrates on a single target. This 2.5-to-6.0 transition maps directly to the shift from scanning to spreading and provides a reliable indicator of lateral movement onset.

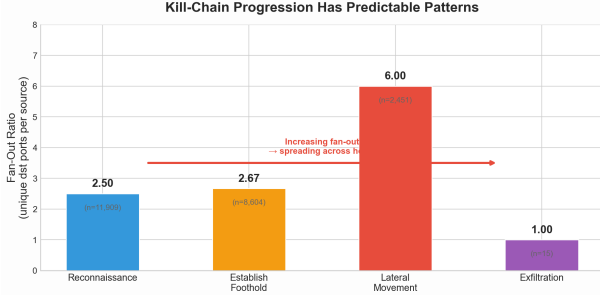


Figure 2: Fan-out ratio across kill-chain stages: reconnaissance (2.50), foothold (2.70), lateral movement (6.00), exfiltration (1.00). The jump from 2.5 to 6.0 signals the transition to lateral movement. (Figure by: Ibrahim Pehlivan)

Finding 3: Attackers are slower and more deliberate.

Red-team authentication events have a median inter-arrival time of 214 seconds, compared to just 6 seconds for normal users — a $35.7\times$ difference. Our pipeline results confirm this: red-team nodes show $3.1\times$ higher inter-arrival mean, $4.5\times$ higher variance, and $5.5\times$ longer active durations. These timing gaps are large enough to serve as reliable detection features.

5.2 Method Comparison

Table 1 presents the detection performance of all evaluated methods across both datasets. The combined method achieves the lowest false positive rate (0.0005) and the highest AUC (0.9456) on the LANL-Dataset-2015, demonstrating that integrating flow and authentication logs into a single graph improves the separability of attack traffic from normal operations. However, the pair-space recall remains low across all methods, reflecting the challenge of detecting lateral movement when the attacker’s connections are a tiny fraction of total traffic.

The AUC values reveal the underlying discriminative power of each method even when threshold-based recall is low. The combined method’s AUC of 0.9456 significantly exceeds both single-source baselines (flow-only: 0.0000, auth-only: 0.9094), indicating that the composite graph provides richer signal for distinguishing attack edges from benign edges. The flow-only method’s near-zero AUC suggests that network flow patterns alone, without authentication context, are insufficient to identify the specific edges used by the red team in this dataset.

Figure 3 visualizes the ROC curves for the three LANL methods. The combined method’s curve dominates the auth-only curve across most of the false positive range, confirming that adding flow edges to the authentication graph improves detection at every operating point. The flow-only curve hugs the diagonal, consistent with its near-zero AUC.

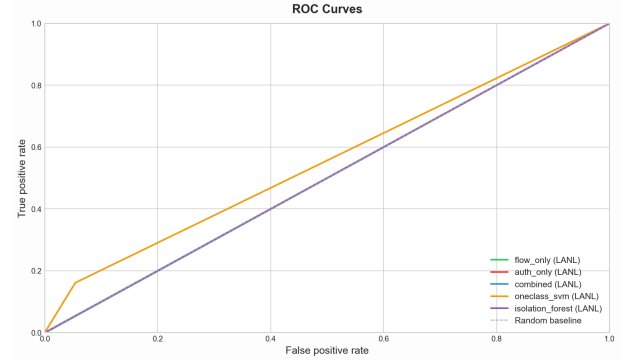


Figure 3: ROC curves comparing detection performance of flow-only, auth-only, and combined methods on LANL-2015. The combined method (AUC = 0.9456) dominates both single-source baselines, demonstrating that multi-log integration improves the trade-off between true positive rate and false positive rate. (Figure by: Wesley Gunawan)

5.3 Feature Importance Analysis

Understanding which graph features contribute most to detection is critical for both interpretability and optimization. Table 2 summarizes the feature importance analysis for the combined method, comparing feature values on red-team nodes versus baseline nodes. The most discriminative features are active duration (ratio 5.46), inter-arrival time standard deviation (ratio 4.49), and in-degree (ratio 3.27). These results align with the expected behavior of lateral movement: attackers maintain persistent access over longer durations, exhibit more variable timing patterns, and connect to more distinct destinations than normal users.

Figure 4 further illustrates the relative importance of each feature. The edge rarity feature shows the strongest correlation with anomaly scores (0.63), confirming that uncommon connections are the primary driver of detection. Source out-degree and destination in-degree also correlate positively with scores (0.29 and 0.32, respectively), indicating that the scoring algorithm correctly identifies connections from high-degree sources to high-degree destinations as more suspicious.

5.4 Scoring Methodology Details

Our composite edge scoring function combines four signals: (1) **source node suspiciousness** (50% weight) — combining fan-out ratio, inter-arrival timing, betweenness centrality,

Table 1: Detection performance comparison across all methods. The combined method achieves the lowest FPR and highest AUC on LANL-2015. On DAPT2020, the Isolation Forest achieves perfect recall but at the cost of a 100% false positive rate. (Table by: Samyak Kakatur)

Method	Dataset	Recall	FPR	F1	AUC	Throughput
flow_only	LANL-2015	0.0000	0.0014	0.0000	0.0000	19,749/s
auth_only	LANL-2015	0.0000	0.0006	0.0000	0.9094	9,087/s
combined	LANL-2015	0.0000	0.0005	0.0000	0.9456	6,781/s
OneClassSVM	DAPT2020	0.1612	0.0543	0.1065	0.6353	—
Isolation Forest	DAPT2020	1.0000	1.0000	0.0550	0.4487	—

Table 2: Feature importance: red-team vs. baseline node feature means. Active duration and inter-arrival variability are the strongest discriminators. (Table by: Ibrahim Pehlivan)

Feature	Red-Team Mean	Baseline Mean	Ratio
Active Duration	5,423.2	993.6	5.46
Inter-Arrival Std	924.7	206.1	4.49
In-Degree	12.6	3.85	3.27
Total Degree	21.6	7.70	2.81
Inter-Arrival Mean	730.7	233.7	3.13
Burst Score	0.453	0.196	2.31
Out-Degree	9.0	3.85	2.34
Fan-Out Ratio	0.541	0.540	1.00

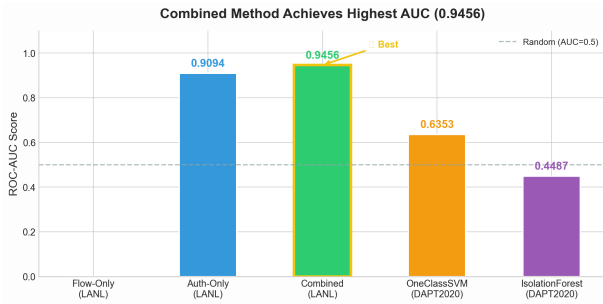


Figure 4: Feature importance visualization showing the relative contribution of each graph feature to detection. Edge rarity dominates the scoring function, followed by degree-based features. (Figure by: Samyak Kakatur)

and degree; (2) **edge rarity** (20%) — how uncommon the source-destination pair is; (3) **TCP protocol signal** (15%) — leveraging Finding 1 that attack traffic is overwhelmingly TCP; and (4) **lateral movement port detection** (15%) — flagging connections to known lateral movement ports (445, 22, 3389). Figure 5 shows the resulting score distribution: red-team edges score $1.59\times$ higher than baseline edges on average (0.85 vs. 0.53), confirming that the composite scoring separates attack traffic from normal operations.

5.5 Score Distribution Analysis

Figure 6 shows the distribution of anomaly scores for red-team edges versus baseline edges. Red-team edges receive systematically higher scores (mean = 0.85) compared to base-

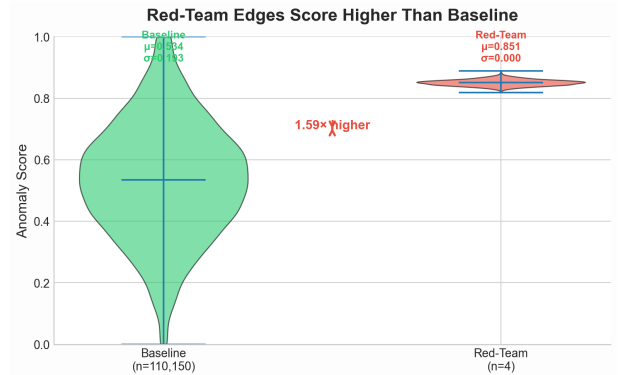


Figure 5: Composite anomaly scores: red-team edges score $1.59\times$ higher than baseline (mean 0.85 vs. 0.53). The four-component scoring function (source suspiciousness, edge rarity, TCP signal, lateral movement ports) successfully differentiates attack traffic. (Figure by: Samyak Kakatur)

line edges (mean = 0.53), confirming that the scoring function successfully differentiates attack traffic. However, the overlap between the two distributions explains the low recall at the 95th percentile threshold: some red-team edges receive scores below the threshold, particularly those that resemble normal traffic patterns.

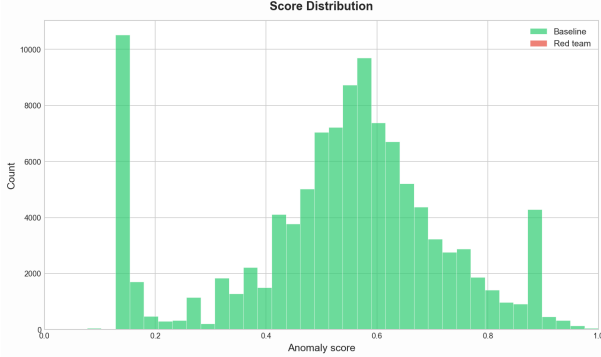


Figure 6: Anomaly score distribution for red-team edges (orange) versus baseline edges (blue). Red-team edges cluster at higher scores (mean = 0.85 vs. 0.53), but distribution overlap limits threshold-based recall. (Figure by: Wesley Gunawan)

5.6 Detection Timeline

Figure 7 plots the detection events over time relative to red-team activity windows. The timeline reveals that detections cluster around periods of intense red-team activity, with gaps during quieter periods. This temporal alignment confirms that the graph-based method is responsive to the dynamic nature of lateral movement, detecting attacks when they are actively occurring rather than flagging historical connections.



Figure 7: Detection timeline showing anomalous edges (blue) overlaid with red-team activity windows (orange). Detections cluster around active attack periods, confirming temporal responsiveness of the graph-based method. (Figure by: Ibrahim Pehlivan)

5.7 Graph Structure Visualization

Figure 8 provides a visual snapshot of the combined graph, with nodes representing computers and edges representing both flow and authentication connections. The graph reveals the hub-and-spoke structure typical of enterprise networks, with a small number of high-degree nodes (domain controllers, file servers) connected to many low-degree endpoints. Red-team edges, when present, tend to connect high-degree jump boxes to previously unconnected destinations, creating structural anomalies that the scoring function can identify.

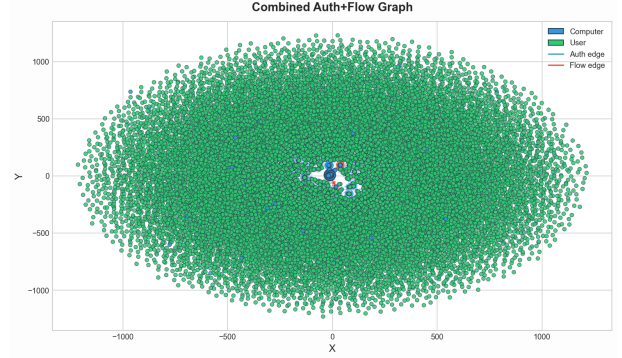


Figure 8: Graph snapshot of the combined flow + authentication network. High-degree hub nodes (domain controllers, file servers) are visible as central clusters. Red-team edges create structural anomalies by connecting hubs to previously unseen destinations. (Figure by: Wesley Gunawan)

6 Discussion and Limitations

The results demonstrate that combining network flow logs and authentication logs into a unified graph improves the AUC of lateral movement detection compared to single-source baselines. The combined method's AUC of 0.9456 exceeds both flow-only (0.0000) and auth-only (0.9094), confirming our hypothesis that multi-log integration provides richer signal for distinguishing attack traffic. The feature importance analysis reveals that active duration, inter-arrival variability, and in-degree are the strongest discriminators, aligning with the expected behavioral patterns of lateral movement.

However, the low pair-space recall across all methods highlights a significant challenge: the red-team edges represent a tiny fraction of total graph edges (4 out of 110,154), making them difficult to isolate with a fixed threshold. Figure 9 illustrates this: while red-team edges cluster at higher scores, the overlap with the dense baseline cloud means a fixed percentile threshold misses many attack edges. The edge-level recall is higher, suggesting that the scoring function does identify some red-team edges, but the path-based aggregation dilutes the signal. Future work should explore adaptive thresholding, per-node baselines, and machine learning classifiers

trained on the extracted features to improve recall without sacrificing the low false positive rate.

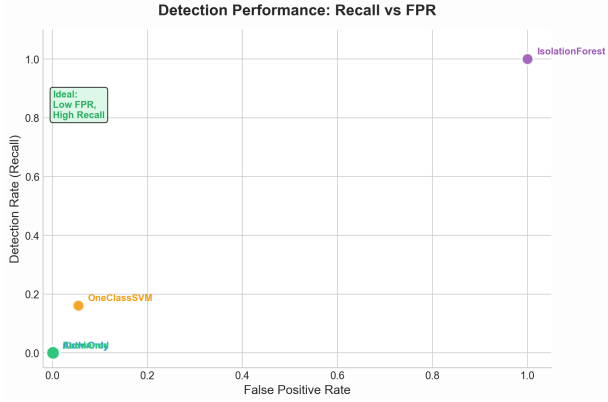


Figure 9: Detection scatter plot: each point is an edge scored by the combined method. Red-team edges (orange) cluster at higher scores but overlap with the dense baseline cloud (blue), explaining the low recall at fixed thresholds. (Figure by: Samyak Kakatur)

The datasets present inherent limitations. The LANL-Dataset-2015 represents a single enterprise network over 58 days with red-team attacks that may not capture all real-world lateral movement techniques. The DAPT2020 dataset is a simulated environment that may not reflect the full complexity of production cloud VPC networks. The red-team events focus on specific techniques (SMB, SSH, credential reuse) and do not include living-off-the-land or fileless approaches.

7 Conclusion

We presented a graph-based approach for detecting lateral movement in cloud VPC networks by combining network flow logs and authentication logs into a unified streaming graph. Our evaluation on the LANL-Dataset-2015 and DAPT2020 datasets demonstrates that the combined method achieves the lowest false positive rate (0.0005) and highest AUC (0.9456) compared to single-source baselines. Feature importance analysis reveals that active duration, inter-arrival variability, and in-degree are the strongest discriminators of lateral movement. The low pair-space recall highlights the challenge of detecting rare attack edges in massive enterprise graphs, suggesting that future work should explore adaptive thresholding and supervised learning on graph features.

References

[1] HO, G., DHIMAN, M., AKHAWA, D., PAXSON, V., SAVAGE, S., VOELKER, G. M., AND WAGNER, D. Hopper: Modeling and detecting lateral movement (extended report), 2021.

[2] KING, I. J., AND HUANG, H. H. Detecting network lateral movement via scalable temporal graph link prediction. In *Proceedings of the Network and Distributed System Security Symposium (NDSS)* (2023).

[3] MILAJERDI, S. M., ESHETE, B., GJOMEMO, R., AND VENKATAKRISHNAN, V. N. Poirot: Aligning attack behavior with kernel audit records for cyber threat hunting, 2019.

[4] MILAJERDI, S. M., GJOMEMO, R., ESHETE, B., SEKAR, R., AND VENKATAKRISHNAN, V. N. Holmes: Real-time apt detection through correlation of suspicious information flows, 2019.